HYDRO RAIN GUARD

Integrated Intelligent Assessment Report

Rainwater Harvesting & Artificial Groundwater Recharge

Field	Value
Project / Owner	श्रीमती सुनीता पाटील
Location	कोल्हापूर, महाराष्ट्र
Coordinates	16.7050, 74.2433
Sampled near grid cell	18.75°, 73.75°
Report generated	26 October 2026, 07:29
Assessment ID	2680FD69
Rainfall source	NOAA PSL CMAP (CPC Merged Analysis of Precipitation)

Executive Summary

Metric	Value	Unit
Annual rooftop runoff	206.6	KL/year
Harvestable volume	176.6	KL/year
Groundwater recharge potential	85.7	KL/year
Recharge feasibility score	70	/ 100
Annual water cost savings	6,181	INR/year
Groundwater recharge value	1,500	INR/year
Estimated installation cost	94,495	INR
Payback period	16.3	Years
10-year ROI	-38.7	%

Rainfall Data & Provenance

Item	Value
Dataset	NOAA PSL CMAP (CPC Merged Analysis of Precipitation)
Method	NOAA PSL CMAP (CPC Merged Analysis of Precipitation), 2.5° grid
Climatology period	1991/01/01 - 2020/12/31
Grid resolution	2.5°
Interpolated from	18.75°, 73.75°
Annual total	1,530 mm

Item	Value
Wettest month	Jul (434 mm)
Driest month	Jan (1 mm)

Note: the annual total above comes straight from the gridded climatology. At 2.5° a cell is roughly 278 km across, so it cannot resolve local orographic gradients — for example the Western Ghats escarpment, where a single cell spans both the high-rainfall crest and the much drier rain-shadow to its east. Where a local station total is known (IMD, state groundwater board, or a rain gauge on site), re-run the assessment with that figure as the annual override: the monthly distribution is kept and only the total is rescaled.

Grid estimate — local override recommended if available

Input Parameters

Parameter	Value
Roof type	RCC
Roof area	150 m²
Occupants	6
Open space area	80 m²
Soil type	LOAMY
Groundwater depth	12.0 m
Annual rainfall	1,530 mm
Runoff coefficient (C)	0.90
Curve number (CN)	98
Soil infiltration rate	12.5 mm/hr

Monthly Rainfall & Runoff

Primary method: Rational / volumetric (Q = C × R × A, CGWB 2007). C = 0.90, CN = 98. The SCS-CN column is reported as a cross-check: run on monthly totals it differs from the Rational method by +7%, which is the expected spread between the two approaches.

Month	Rainfall (mm)	Runoff (L)	SCS-CN (L)	Savings (INR)
Jan	1.3	176	2	5
Feb	1.3	176	2	5
Mar	6.7	904	444	27
Apr	15.7	2,120	1,625	63
May	54.3	7,330	7,281	219
Jun	371.4	50,139	54,788	1,500
Jul	433.9	58,576	64,161	1,752
Aug	314.2	42,417	46,210	1,269
Sep	199.6	26,946	29,027	806
Oct	107.8	14,553	15,273	435
Nov	18.9	2,552	2,077	76
Dec	5.2	702	278	21
ANNUAL	**1,530**	**206,577**	**221,166**	**6,181**

Groundwater Recharge Assessment

Parameter	Value
Aquifer type	Semi-Confined Aquifer
Groundwater depth	12.0 m
Soil infiltration rate	12.5 mm/hr
Annual recharge potential	85.7 KL/year
Daily recharge rate	235 L/day
Recharge feasibility score	70 / 100
Aquifer permeability	Moderate (10-15 mm/hr)
Groundwater quality	Potable - requires testing
Seasonal fluctuation	6.1 m
Aquifer depth	12.0 m
Transmissivity	125.0 m^2/day
Storage coefficient	0.150

Recommended Structures

Ranked by Analytic Hierarchy Process (Saaty, 1980) — a weighted, consistency-checked multi-criteria score. This is a deterministic rule-based model, not machine learning.

Criterion	Weight
Harvestable yield	41.7%
Annual rainfall	26.3%
Soil infiltration	16.0%
Open space	9.8%
Groundwater depth	6.2%
Consistency ratio (CR)	0.015

CR = 0.015 (λmax = 5.068). Saaty's acceptability threshold is 0.10, so the pairwise judgements are consistent.

#1 Rooftop Storage Tank — 100 / 100

Yield	Rainfall	Soil	Space	Depth
1.00	1.00	1.00	1.00	1.00

A dedicated storage tank collects and stores harvested rooftop runoff for direct use in toilets, gardening, and non-potable purposes. Most cost-effective first step for any building.

Dimensions — volume (m^3): 29.3 | diameter m: 4.32 | depth m: 2.0 | material: RCC/PVC

Installation — Install at ground level or underground. Connect to rooftop downpipes via first-flush diverter. Include overflow pipe to recharge pit.

#2 Recharge Well / Borewell Recharge — 98 / 100

Yield	Rainfall	Soil	Space	Depth
1.00	1.00	1.00	1.00	0.70

A dedicated recharge well (200–300mm diameter, 10–30m deep) channels treated rooftop runoff directly into deeper aquifers. Ideal where surface soil has low permeability but deep aquifers are present.

Dimensions — diameter mm: 250 | depth m: 20 | casing type: MS perforated pipe

Installation — Drill 200–300mm bore to aquifer depth. Install casing with perforations. Connect via first-flush diverter -> settling tank -> filter media -> well. Annual cleaning required.

#3 Recharge Pit — 87 / 100

Yield	Rainfall	Soil	Space	Depth
1.00	1.00	0.21	1.00	1.00

A recharge pit (1.5m × 1.5m × 2m) filled with gravel and sand filters allows harvested water to percolate directly into the groundwater table. Highly effective for shallow aquifers.

Dimensions — length m: 1.5 | width m: 1.5 | depth m: 2.0 | filter layers: 0.5m gravel + 1.0m stone + 0.5m sand

Installation — Excavate 2m deep, fill with 0.5m gravel at bottom, 1m stone/brick, 0.5m sand. Install inlet pipe from tank overflow. Provide removable top cover for cleaning.

Cost-Benefit Analysis

Item	Amount (INR)
Total installation cost	94,495
Storage structures	25,000

Item	Amount (INR)
Recharge structures	45,000
Plumbing & conveyance	24,495
Annual water savings (volume)	176.6 KL
Annual cost savings	6,181
Groundwater recharge value	1,500
Annual maintenance (2%)	1,890
Net annual savings	5,790
Water tariff assumption	35.0 / KL
Payback period	16.3 years
10-year ROI	-38.7%

Methods & Assumptions

Rainfall — gridded monthly climatology, bilinearly interpolated between the four surrounding grid cells; no-data cells are excluded from the interpolation. Runoff — Rational / volumetric method (Q = C × R × A) with the SCS Curve Number method reported as a cross-check; both are standard, published methods, not fitted models. First flush — 2 mm discarded per rain event, assumed 100 events per year. Recharge — infiltration limited by the soil's saturated hydraulic conductivity and capped at the rainfall actually falling on the open area. Economics — capital cost from unit rates, 2% of capital per year for maintenance, savings valued at the water tariff stated above; no discount rate is applied to the payback figure. Structure ranking — Analytic Hierarchy Process, a deterministic weighted scoring method; it is not machine learning and uses no training data.

Screening-level estimates only. Verify against a site-specific hydrogeological investigation before construction.

All figures are engineering estimates for pre-feasibility screening. Verify against site-specific investigation before construction.